# Work done

National 5 · Engineering contexts & challenges · Answers

## Answers and marking guidance

Working is shown for each answer. Other valid methods should be credited.

**1 Calculate the work done.**
- $E_w = F \times d = 400 \times 25$
- $E_w = 10\,000$ J (10 kJ)

**2a) 0.5 kg**
- $F = 0.5 \times 9.8 = 4.9$ N
- $E_w = 4.9 \times 2 = 9.8$ J

**2b) 2.1 kg**
- $F = 2.1 \times 9.8 = 20.58$ N
- $E_w = 20.58 \times 2 = 41$ J

**2c) 3.2 kg**
- $F = 3.2 \times 9.8 = 31.36$ N
- $E_w = 31.36 \times 2 = 63$ J

**2d) 3.5 kg**
- $F = 3.5 \times 9.8 = 34.3$ N
- $E_w = 34.3 \times 2 = 69$ J

**3 A force of 150 N pulls a 50 kg bag of sand 20 m along the ground. Calculate the work done.**
- $E_w = F \times d = 150 \times 20 = 3000$ J
- The mass is not needed: the force applied is given.

**4a) Calculate the minimum work the truck must do.**
- $F = 740 \times 9.8 = 7252$ N
- $E_w = 7252 \times 2 = 14\,504$ J = 15 kJ

**4b) Suggest why the actual work done will be greater than this.**
- Energy is also needed to lift the forks and mechanism, and to overcome friction in the machine.

**5 A gardener pushes a lawnmower 42 m with an average force of 230 N. Calculate the work done.**
- $E_w = 230 \times 42 = 9660$ J = 9.7 kJ

**6 A car is brought to rest over 24 m by a frictional force of 980 N. Calculate the work done.**
- $E_w = 980 \times 24 = 23\,520$ J = 24 kJ

**7 A parent pushes a pram 768 m. The total work done is 210 kJ. Calculate the average force applied.**
- $F = E_w \div d = 210\,000 \div 768$
- $F = 273$ N = 270 N

**8 A car tows a caravan 12 km. The total work done is 30 MJ. Calculate the average force.**
- $F = 30\,000\,000 \div 12\,000$
- $F = 2500$ N (2.5 kN)

**9 A tug boat does 205 MJ of work by exerting an average force of 15 kN on an oil rig. Calculate how far the rig is towed.**
- $d = E_w \div F = 205\,000\,000 \div 15\,000$
- $d = 13\,667$ m = 14 km

**10a) Calculate the weight of the person.**

- $W = mg = 50 \times 9.8 = 490$ N

**10b) Calculate the work done.**

- $E_w = 490 \times 8 = 3920$ J = 3.9 kJ